# HW5 Question 2

Task 1: ( $\underline{w}$ means column vector, e.g. $\underline{w} = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix}$ )

1. Given posterior $\propto$ likelihood $\times$ prior

$$\Rightarrow \text{posterior} \propto p(y \mid X, \underline{w}) \times p(\underline{w} \mid X, y)$$

$$\Leftrightarrow \text{posterior} = k \cdot \exp\left(-\frac{1}{2\sigma^2}(y - X\underline{w})^T (y - X\underline{w})\right) \cdot p(\underline{w})$$

Define log posterior as $\mathcal{L}_p(\underline{w})$

$$\mathcal{L}_p(\underline{w}) = \log(\text{posterior}) = \log k + \log p(\underline{w}) - \frac{1}{2\sigma^2}(y - X\underline{w})^T (y - X\underline{w})$$

By definition, the prior over $\underline{w}$ is

$$p(\underline{w}) = \mathcal{N}(\underline{0}, \tau^2 I) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2\tau^2} w^T w\right)$$

$$\Rightarrow \mathcal{L}_p(\underline{w}) = \log k - \frac{1}{2}\log(2\pi) - \frac{1}{2\tau^2}\underline{w}^T \underline{w} - \frac{1}{2\sigma^2}(y - X\underline{w})^T (y - X\underline{w})$$

$$= \log k - \frac{1}{2}\log(2\pi) - \frac{1}{2\tau^2} w^T w - \frac{1}{2\sigma^2}\left(y^T y - y^T X\underline{w} - (X\underline{w})^T y + (X\underline{w})^T X\underline{w}\right)$$

$$= \log k - \frac{1}{2}\log(2\pi) - \frac{1}{2\tau^2} w^T w - \frac{1}{2\sigma^2}\left(y^T y - 2 w^T X^T y + \underline{w}^T X^T X \underline{w}\right)$$

Define $\log k - \frac{1}{2}\log(2\pi) - \frac{1}{2\sigma^2} y^T y \equiv C_0$, $I$ is $N \times N$

$$\mathcal{L}_p(\underline{w}) = C_0 - \underline{w}^T\left(\frac{1}{2\tau^2}I + \frac{1}{2\sigma^2}X^TX\right)\underline{w} + \frac{1}{\sigma^2}\underline{w}^TX^Ty$$

$$= C_0 - \frac{1}{2}\underline{w}^T\left(\frac{1}{\tau^2}I + \frac{1}{\sigma^2}X^TX\right)\underline{w} + \frac{1}{\sigma^2}\underline{w}^TX^Ty$$

2. Let $\Sigma^{-1} = \frac{1}{\tau^2}I + \frac{1}{\sigma^2}X^TX$

$$\mathcal{L}_p(\underline{w}) = C_0 - \frac{1}{2}\underline{w}^T\Sigma^{-1}\underline{w} + 2\cdot\left(\frac{1}{2}\underline{w}^T\Sigma^{-1}\right)\cdot\left(\frac{1}{\sigma^2}\Sigma X^Ty\right) - C_1$$

Let $\underline{\mu} = \frac{1}{\sigma^2}\Sigma X^Ty$, $C_0 - C_1 = \text{Const}$

$$\mathcal{L}_p(\underline{w}) = -\frac{1}{2}(\underline{w}-\mu)^T\Sigma^{-1}(\underline{w}-\mu) + \text{Const}.$$

Hence we obtain the form of a Gaussian,

3. $\Rightarrow$ where $\mu = \frac{1}{\sigma^2}\left(\frac{1}{\tau^2}I + \frac{1}{\sigma^2}X^TX\right)^{-1}X^Ty$ (mean) ①

$\Sigma = \left(\frac{1}{\tau^2}I + \frac{1}{\sigma^2}X^TX\right)^{-1}$ (Covariance) ②

$\Rightarrow$ the posterer Gaussian.

$$\underline{w} \mid X, y \sim \mathcal{N}\left(\frac{1}{\sigma^2}\left(\frac{1}{\tau^2}I + \frac{1}{\sigma^2}X^TX\right)^{-1}X^Ty, \left(\frac{1}{\tau^2}I + \frac{1}{\sigma^2}X^TX\right)^{-1}\right)$$

(To be continued)

Task 2

1. The posterior is $\underline{w} \mid X, \underline{y} \sim \mathcal{N}(\underline{\mu}, \Sigma)$

Hence $p(\underline{w} \mid X, \underline{y}) \propto \exp\left(-\frac{1}{2\Sigma^2}(\underline{w}-\underline{\mu})^T(\underline{w}-\underline{\mu})\right)$

2. ▲ Suppose we have $y_{new} = w^T \phi(\underline{x}_{new}) + \varepsilon$, where $\varepsilon \sim \mathcal{N}(0, \sigma^2)$

Here, $\phi(x)$ is one (any) column of the design matrix $X$. Here we assume $\phi(\underline{x})$ is just $\underline{x}$, $\phi(\underline{x}_{new}) = \underline{x}_{new}$

$$p(y_{new} \mid \underline{x}_{new}, \underline{w}) \propto \exp\left(-\frac{1}{2\sigma^2}(y - w^T\underline{x}_{new})^T(y - w^T\underline{x}_{new})\right)$$
$$= \mathcal{N}(w^T x_{new}, \sigma^2)$$

$$\Rightarrow p(y_{new} \mid \underline{x}_{new}, X, y) = \int p(y_{new} \mid \underline{x}_{new}, w)\, p(\underline{w} \mid X, \underline{y})\, d\underline{w}$$

$$\propto \int \exp\left(-\frac{1}{2\Sigma^2}(\underline{w}-\underline{\mu})^T(\underline{w}-\underline{\mu}) - \frac{1}{2\sigma^2}(y - w^T\underline{x}_{new})^T(y - w^T\underline{x}_{new})\right) d\underline{w}$$

This suggests that it is a linear combination of two Gaussian random variables $y_{new}$ and $w$.

Hence $\mu_{new} = E[w^T \phi(\underline{x}_{new})] = \mu^T \phi(\underline{x}_{new})$ (new mean)

$$\sigma_{new} = Var[w^T \underline{x}_{new}] + \sigma^2$$
$$= \phi(\underline{x}_{new})^T \Sigma\, \phi(\underline{x}_{new}) + \sigma^2 \text{ (new variance)}$$

where $\mu$ and $\Sigma$ were given by ①, ② in the previous task.

▲ And $\phi(\underline{x})$ is the linear model for

$$y_{new} = w^T \phi(\underline{x}_{new}) + \varepsilon$$

$\Rightarrow$ Taking in $\phi(\underline{x}) = \underline{x}$, the mean $\mu_{new} = \mu^T \underline{x}_{new}$ ($\mu$ given by ①)

the variance $\sigma_{new} = \underline{x}_{new}^T \Sigma\, \underline{x}_{new} + \sigma^2$ ($\Sigma$ given by ②)